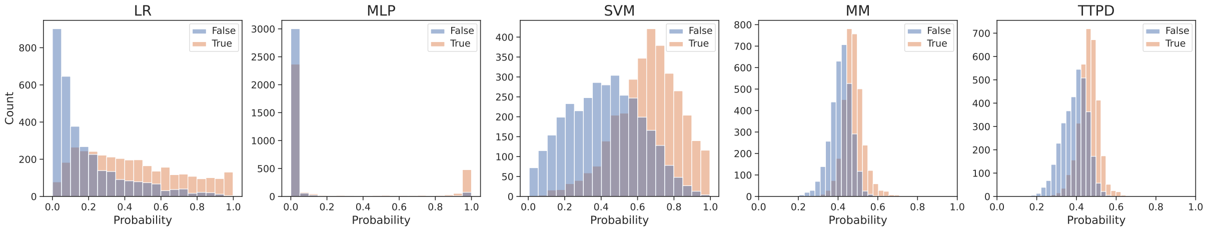


Figure 9: Output distributions of LR, MLP, SVM, MM and TTPD probes on the BoolQ dataset under the "with options" setting. The target LLM is Llama-3.1-8B.

Topic	**Description**	**Example statement**
`animal_class`	The class of a specific animal species.	The salmon is a fish.
`cities`	Locations of world cities.	The city of Krasnodar is in Russia.
`element_symb`	Chemical elements and their abbreviations.	Thallium has the symbol Tl.
`facts`	Diverse scientific facts.	The Earth's atmosphere protects us from harmful radiation from the sun.
`inventors`	Home countries of inventors.	The inventor Edwin Herbert Hall lived in the U.S.
`sp_en_trans`	Translations of Spanish words to English.	The Spanish word 'con' means 'to speak'.

Table 1: Summary of topic-specific factual statement datasets.

E Metric Details

E.1 Expected Calibration Error (ECE)

For ECE, we first sort the probabilities predicted by a probe and split them into N equal-sized bins. In this paper we let $N = 10$, which is common in literature evaluating calibration. For each bin, we calculate the mean probability (x_i) and the fraction of truthful predictions (y_i). ECE is then computed following the formula below:

$$\text{ECE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - x_i|. \tag{1}$$

E.2 Brier Score (BS)

The Brier Score measures the difference between the actual correctness and the confidence score through point-wise mean squared error. Its formulation is as follows:

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N} (p_i - y_i)^2, \tag{2}$$

where p_i is the confidence reported by the probe and $y_i \in \{0, 1\}$ is the ground truth label. When a predictor is always making inconfident random predictions, i.e. $p_i = 0.5 (i = 1, 2, ...N)$, it results in a chance Brier score of 0.25.

F Experiment Details and More Results

In this section, we elaborate on the detailed setups of the experiments in Section 4. Furthermore, as we only use the Llama family of models in the body of the paper, in this section we also demonstrate results on a model of the Mistral family, Mistral-7B-v0.1 (Jiang et al., 2023).

F.1 Computational and Storage Resources

All of our experiments are completed on three A6000 (48GB) GPUs. For most LLMs except Llama-3.1-70B(-Instruct), only one GPU will suffice. However, the storage for activations across all datasets and models would take ~1TB disk space. Therefore we recommend modifying the code and only keep the necessary activations on disk. Furthermore, in order to accommodate large LLMs such as Llama-3.1-70B(-Instruct) into our GPUs when gathering hidden activations, we use `float8` quantization with the `optimum-quanto`[3] library.

[3] `https://github.com/huggingface/optimum-quanto`

F.2 Selecting Layer

In Section 4.1 we discuss the selection of decoder layer residual stream to extract truth direction from. We present a criteria based on the ratio of between-class variance to within-class variance. However, due to limitation of space we only present results for Llama-2-7B and Llama-3.1-8B in Section 4.1. Here we replicate these approaches on more models.

The data used for plotting is the collection of both affirmative and negative atomic statements covering all the six topics, as well as their logical conjunctions and disjunctions. The curve for each topic consists of the four variations of statements, which results in six curves for each model.

We summarize the plots in Figure 10, which covers eight models: Llama-2-7B(-Chat), Llama-2-13B(-Chat), Llama-3.1-8B(-Instruct), Llama-3.1-70B(-Instruct), Mistral-7B(-Instruct)-v0.1. Their optimal layers are 12(13), 13(13), 12(13), 33(33), 13(13), respectively.

F.3 Consistency of Truth Direction

We present results on Mistral-7B-v0.1 in Figure 11. It is evident that the truthfulness probes generalize across negation on four topics, barely generalizing on `facts` topic. Comparing these results with those of Figure 3, we notice that the performance of probes for Mistral-7B-v0.1 is comparable to that of probes for Llama-2-13B-Chat and Llama-3.1-8B. This aligns with the observation that the general capability of Mistral-7B-v0.1 lies between that of Llama-2-13B-Chat and Llama-3.1-8B.

F.4 Logical Conjunction/Disjunction

Full results on logical conjunctions and disjunctions are shown in Figure 12 and Figure 13 respectively. A similar scaling trend could be observed as in Figure 3, where the classification accuracy of the probes is positively correlated with the target LLM's general capability.

The results for Mistral-7B-v0.1 is shown in Figure 14. The truthfulness probes generalize from atomic factual statements to both logical conjunctions and disjunctions.

F.5 Question Answering

F.5.1 Details on Experiment Setup

MMLU. We arrange three setups for the QA task, and we demonstrate the prompt template for zero-shot setting using an actual example from the MMLU dataset. The few-shot prompts are trivially extended from the zero-shot prompt, with exemplars separated by two newlines ("\n\n"). Few-shot exemplars are randomly selected from the development split.

```
Question: What was GDP per capita in the
     United States in 1850 when
    adjusting for inflation and PPP in
    2011 prices?
Options:
A. About $300
B. About $3k
C. About $8k
D. About $15k
Answer: B
```

TriviaQA. For TriviaQA (Joshi et al., 2017) we only use few-shot prompting – 5-shot and 20-shot – to ensure that the LLM always generates short-form answers. We use normalized answers in the exemplars.

```
Question: Where in England was Dame Judi
     Dench born?
Answer: york
```

F.5.2 More results

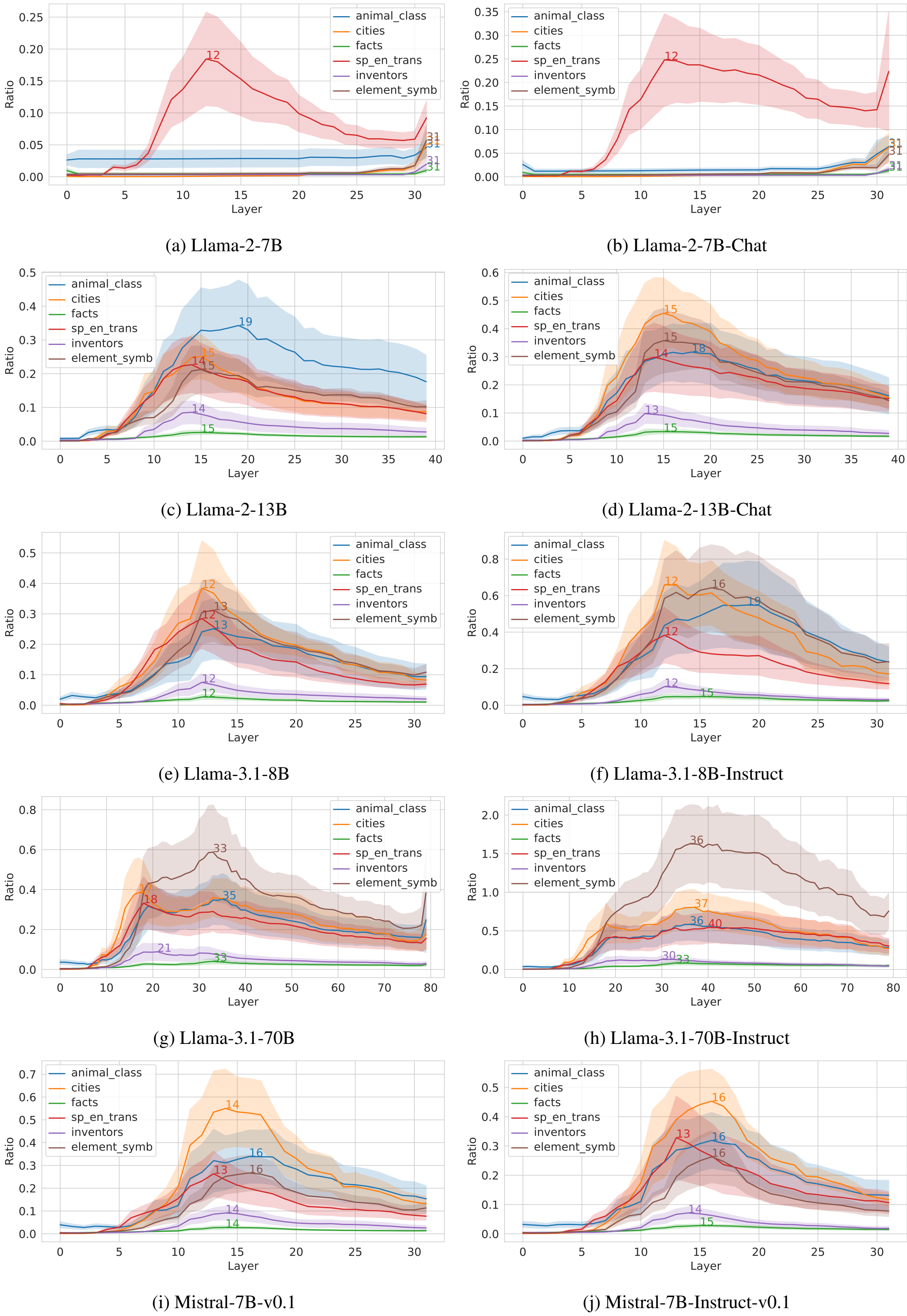
The results for Mistral-7B-v0.1 is shown in Figure 15. The general behavior of truthfulness probes is similar to that in Figure 5. The classification accuracy improves in response to few-shot prompting and improves when more in-context exemplars are provided in the prompt. Meanwhile, calibration only improves in the case of the SVM probe on TriviaQA dataset.

F.6 Contextual Knowledge

F.6.1 Details on Experiment Setup

SciQ. For the SciQ (Welbl et al., 2017) benchmark, we arrange three setups, including "zero-shot", "TTT" and "TTF". We only demonstrate the zero-shot prompt as the few-shot prompts can be trivially extended from it. In the in-context setups, the exemplars are randomly selected from the training split.

```
Context: <context>
Question: Compounds that are capable of
    accepting electrons, such as o 2 or
    f2, are called what?
Options:
A. Oxygen
B. residues
C. antioxidants
D. oxidants
Answer: D
```



(a) Llama-2-7B

(b) Llama-2-7B-Chat

(c) Llama-2-13B

(d) Llama-2-13B-Chat

(e) Llama-3.1-8B

(f) Llama-3.1-8B-Instruct

(g) Llama-3.1-70B

(h) Llama-3.1-70B-Instruct

(i) Mistral-7B-v0.1

(j) Mistral-7B-Instruct-v0.1

Figure 10: Plot of the ratio of between-class variance to within-class variance for a series of models. The shaded regions denote standard error.